

Week – 2

```

Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (book_id) REFERENCES Books(book_id));
ERROR 1046 (3D000): No database selected
mysql> USE bookflow_db
Database changed
mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 1, 1, '2025-01-05'),
-> (2, 2, 2, '2025-01-08'),
-> (3, 3, 3, '2025-01-10'),
-> (4, 1, 2, '2025-02-01'),
-> (5, 2, 1, '2025-02-05'),
-> (6, 3, 2, '2025-02-12'),
-> (7, 1, 3, '2025-03-01'),
-> (8, 2, 3, '2025-03-07'),
-> (9, 3, 1, '2025-03-15'),
-> (10, 1, 1, '2025-04-01');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 1 | 1 | 2025-01-05 |
| 2 | 2 | 2 | 2025-01-08 |
| 3 | 3 | 3 | 2025-01-10 |
| 4 | 1 | 2 | 2025-02-01 |

```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	1	1	2025-01-05
2	2	2	2025-01-08
3	3	3	2025-01-10
4	1	2	2025-02-01
5	2	1	2025-02-05
6	3	2	2025-02-12
7	1	3	2025-03-01
8	2	3	2025-03-07
9	3	1	2025-03-15
10	1	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m ON l.member_id = m.member_id  
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
Anil Kumar	The Alchemist
Anil Kumar	Clean Code
Anil Kumar	Atomic Habits
Anil Kumar	The Alchemist
Priya Sharma	Clean Code
Priya Sharma	The Alchemist
Priya Sharma	Atomic Habits
Ravi Verma	Atomic Habits
Ravi Verma	Clean Code
Ravi Verma	The Alchemist

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books  
-> FROM Books  
-> GROUP BY published_year  
-> ORDER BY published_year;
```

published_year	Total_Books
1988	1
2008	1
2018	1

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Donation_History (  
-> donation_id INT PRIMARY KEY,  
-> book_id INT,  
-> donor_name VARCHAR(100),
```

```
mysql> CREATE TABLE Donation_History (  
-> donation_id INT PRIMARY KEY,  
-> book_id INT,  
-> donor_name VARCHAR(100),  
-> donation_date DATE,  
-> FOREIGN KEY (book_id) REFERENCES Books(book_id));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)  
-> VALUES (1, 4, 'Raj Kumar', CURDATE());  
Query OK, 1 row affected (0.01 sec)  
  
mysql> COMMIT;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';  
Empty set (0.00 sec)  
  
mysql> |
```